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ORIGINAL INVESTIGATION

Identification of Sensitive Measures of Recovery After External Load From Football Match Play

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Objective measures of recovery from football match play could be useful for assessing athletes' readiness to train, if sensitive to preceding match load. ***Purpose:*** To identify the sensitivity of countermovement-jump (CMJ) performance and concentration of salivary testosterone and cortisol relative to elite football match load. ***Methods:*** CMJ performance and salivary hormones were measured in 18 elite football players before (27, 1 h) and after (0.5, 18, 42, 66, 90 h) 3 consecutive matches. Match load was determined via accelerometer-derived PlayerLoad and divided into tertiles. Sensitivity of CMJ performance and hormone concentrations to match load was quantified with *t* statistics and magnitude-based inferences (change in mean as % ± 90% confidence interval) derived with a linear mixed model. ***Results:*** Jump height was reduced in medium and high load at 0.5 h (10% ± 7% and 16% ± 8%) and 18 h (7% ± 4% and 9% ± 5%) postmatch. There was a 12% ± 7% reduction in ratio of flight time to contraction time (FT:CT) in high load at 0.5 h post, with reductions in medium and high load at 18 h. Reductions in FT:CT persisted at later postmatch time points than changes in jump height. Increased cortisol (range 55–165%) and testosterone (range 17–20%) were observed in all match loads at 0.5 h post, with individual variability thereafter. ***Conclusions:*** Measures of CMJ performance and hormonal concentrations were sensitive to levels of A League football match load. Although jump height was reduced immediately postmatch, FT:CT provided a more sensitive measure of recovery. Football match play induces an acute hormonal response with substantial individual variability thereafter.

Keywords: neuromuscular fatigue, testosterone, cortisol, monitoring

Monitoring the external load performed by an athlete and the response to that load is critical to determine when to apply the next training stimulus.[1] Velocity and distance metrics are commonly reported measures of external load, yet they may underestimate the load in sports such as football where frequent changes of velocity and direction occur.[2] Triaxial accelerometers are used to measure global exercise load.[3,4] PlayerLoad, representing the instantaneous rate of change of the mediolateral, anteroposterior, and vertical accelerometer vectors, is a valid and reliable measure of external load in team sports.[3] PlayerLoad discriminates between levels of performance and is modified due to fatigue.[3,4] PlayerLoad thus may provide a more complete measure of football load than traditional distance and velocity metrics. A commonly quantified response to external load is neuromuscular fatigue.[5–7] Although gold-standard assessment of neuromuscular fatigue requires magnetic or electrical nerve stimulation, this approach is impractical in the applied sport setting.[8] To overcome this limitation, neuromuscular fatigue can be quantified via countermovement-jump (CMJ) performance.[5–7] Of the variables measured via CMJ, some display high ecological validity as they respond negatively to Australian Rules football match play and also impact subsequent match exercise intensity.[5,9,10] The CMJ response to football match play has generally been measured via jump height.[6,11,12] The ratio of flight time to contraction time (FT:CT) may provide a more precise quantification of neuromuscular fatigue in team-sport athletes than jump height because it reflects a change in movement strategy.[6] Therefore jump-height measurement may underestimate the magnitude of neuromuscular fatigue.[5,7,13] Note that altered CMJ performance should be sensitive to preceding exercise load if it is a valid measure of fatigue in the sporting context, but this concept has received only limited attention in the literature.[14]

The hormonal response to match play (testosterone [T] and cortisol [C] concentration in particular) is also commonly reported.[1] T is an anabolic hormone with an important role in promoting protein synthesis, while C is a catabolic hormone important in metabolism.[12,15] The concentrations of T and C are often combined as the T:C ratio, which represents the anabolic–catabolic balance.[12,15] Australian Rules football match play resulted in an acute increase in C with little impact on T, while the T:C ratio was reduced for up to 48 hours postmatch, suggestive of a somewhat catabolic state for this relatively short period.[12]

Despite support for the use of both CMJ performance and hormonal response to assess the impact of football match play, the influence of preceding match load on the magnitude of the postmatch response is not well understood. Therefore, the aim of this research was to determine the impact of football match load on CMJ performance, T, C, and the T:C ratio.

Methods

Subjects

Data (expressed as mean ± SD) were collected from 18 elite football players (age 23.3 ± 4.1 y, height 180.0 ± 10.0 cm, mass 75.7 ± 4.4 kg) from 1 Australian A League club across 3 consecutive preseason Football Federation Australia matches.

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Design

A maximal CMJ and a saliva sample were collected 27 hours and 1 hour prematch and 0.5, 18, 42, 66, and 90 hours postmatch. Athletes were familiarized with CMJ and saliva-collection techniques during a 4-week period before the first match. The research was approved by the Victoria University human research ethics committee, with informed consent obtained before commencement.

Methodology

Athletes completed a standardized 2-minute dynamic warm-up with 3 practice jumps before a maximal CMJ.[5] The CMJ was performed on a force plate (400 Series Platform Plate; Fitness Technology, Adelaide, Australia) connected to manufacturer-supplied software (Ballistic Measurement System; Fitness Technology, Adelaide, Australia). Participants performed 1 CMJ for maximum height with a self-selected depth while maintaining hands in position on the hips.[5] Jump height (m), peak velocity (m/s), relative peak and mean power (W/kg), relative peak force (N/kg), contraction time (s), and FT:CT ratio were analyzed.[9]

Athletes maintained their usual dietitian-prescribed diet throughout the testing period and were instructed to ingest only water in the 60 minutes before providing a saliva sample.[5] To account for diurnal variation, collection of T and C samples generally occurred between 9:00 and 9:30 AM, with the exception of 1 hour prematch and 0.5 hour postmatch. The 1-hour-prematch samples for games 1 and 2 were collected between 12:45 and 1:15 PM and for game 3 between 2:20 and 2:50 PM. The 0.5-hour-postgame samples were collected between 4:00 and 4:30 PM for games 1 and 2 and between 5:30 and 6:00 PM after game 3. Athletes provided 2 mL of unstimulated passive drool directly into a plastic tube.[5] Samples were frozen at –80°C for later analysis. Duplicate enzyme-linked immunosorbent assay (Salimetrics, State College, PA, USA) using a microplate reader (SpectraMax 190, Molecular Devices, Sunnyvale, CA, USA) determined C (µg/dL) and T (pg/mL) concentration.[5] Accelerometer-derived (100-Hz) PlayerLoad expressed in arbitrary units (au) accounted for match load. Match load was divided into 3 levels: low load (0–499 au), medium load (500–1000 au), and high load (>1000 au). The distribution of PlayerLoad from subsequent training relative to low, medium, and high loads from the match is displayed in Figure 1. The low-load group trained at 18 hours postmatch following data collection after games 1 and 2. After matches 1, 2, and 3, all groups trained following data collection at 42 and 66 hours after matches 1 and 2 only. After match 1, 90 hours postmatch was a day off. There was a crossover with 90 hours after match 2 and match 3, and this was accounted for in the statistical model. Regardless of match load, all players trained at 90 hours after match 3.

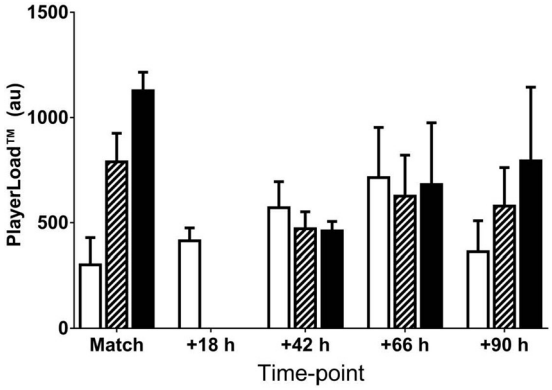

Figure 1 — Distribution of match and training PlayerLoad (au) for low-, medium-, and high-load groups. Data are presented as group means ± SD. Low load is depicted in the unfilled columns, medium load in the striped columns, and high load in the solid columns.

Statistical Analysis

Separate analyses were performed for each of the measures derived from the CMJ and the saliva samples. Each analysis was performed with the same mixed model using the general linear mixed-model procedure (Proc Mixed) in the Statistical Analysis System (version 9.4, SAS Institute, Cary, NC). The fixed effects in the model estimated means for each time point, each game, and each of the 3 levels of match load. Athlete identity and its interaction with match identity were included as random effects to account for repeated measurement on athletes within and between matches, while a different residual error was specified for each of the 7 time points to allow for individual differences in the response to the match load. Jump height and peak velocity had the smallest residual error on the last time point (90 h), which was therefore used as the reference level for estimating changes in performance between time points.

All measures were log-transformed before analysis and then back-transformed to express the changes in percentages. To compare the sensitivity of the measures to the effects of match load, the changes were expressed as *t* statistics, which represent a signal-to-noise ratio. The *t* statistics were provided directly by the mixed-modeling procedure. The changes were also expressed in standardized units for assessment of magnitude. Standardization was performed by dividing the change score of the log-transformed measure by the between-subjects SD derived from the random effects for athlete identity and its interaction with match identity; this between-subjects SD represents the differences between athletes at any given time point free of residual (measurement) error. Uncertainty in the changes was expressed as 90% confidence intervals and interpreted via the nonclinical magnitude-based-inference approach.[16] Standardized changes (ES) of 0.20, 0.60, 1.20, 2.0, and 4.0 were thresholds for small, moderate, large, very large, and extremely large effects, respectively. When the confidence interval for a change included small positive and negative effects, the change was deemed unclear. For clear effects, the likelihood that the true effect was substantial was indicated with the following scale: possibly (25–75%), likely (75–<95%), very likely (95–99.5%), and most likely (>99.5%).

Results

Selecting the Most Sensitive CMJ Measure

Although there was variability relative to match load and time points, jump height, FT:CT, and peak velocity displayed the largest ES and *t*-statistic values (ranges of 0.2–3.0, 0.4–3.9, and 0.0–3.0, respectively) compared with baseline (90 h post). Peak velocity and jump height displayed the same pattern of response; however, the standardized effects for jump height were nearly twice that of peak velocity. As jump height is a commonly reported measure of CMJ performance, it was selected ahead of peak velocity as the variable for further analysis.

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Magnitude of Change Compared With Baseline

Mean jump height and FT:CT response to match load are displayed in Figures 2 and 3, with the change in mean from baseline displayed in Table 1. At 0.5 hour and 18 hours post there was a moderate and small likely reduction (10%, ±7%, and 7%, ±4%) in jump height for medium load. High load displayed a very likely (large and moderate effects) reduction in jump height (16%, ±8%, and 9%, ±5%) at the same time points. At 42 hours, the change in jump height was unclear across all levels of load. High load had a very likely 12%,

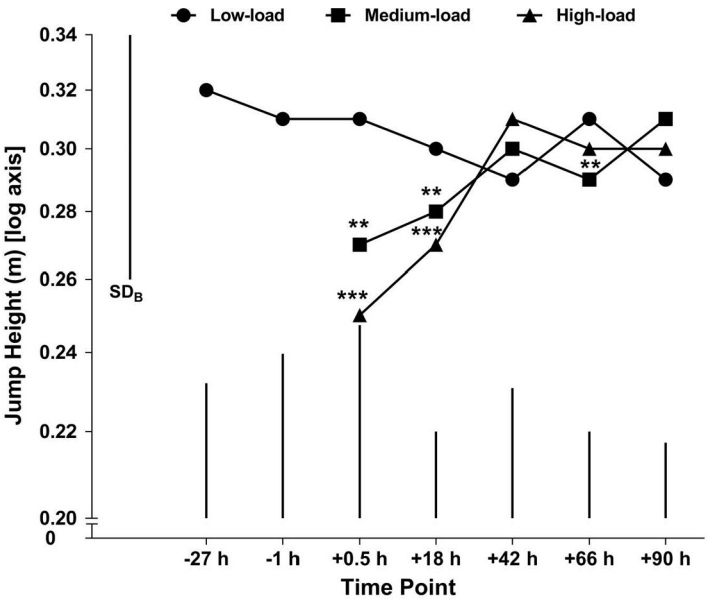

Figure 2 — Mean jump height from 27 hours pre to 90 hours post of low, medium, and high match loads. Mean values are displayed in log scale with the overall between-subjects SD shown by the bar labeled SD_B. The single bars underneath each time point represent the within-subject SD specific to that time point. Changes exceeding the effect size (0.2) with the qualitative descriptor of likely ** (75–<95%), very likely *** (95–99.5%), and most likely **** (>99.5%) are displayed.

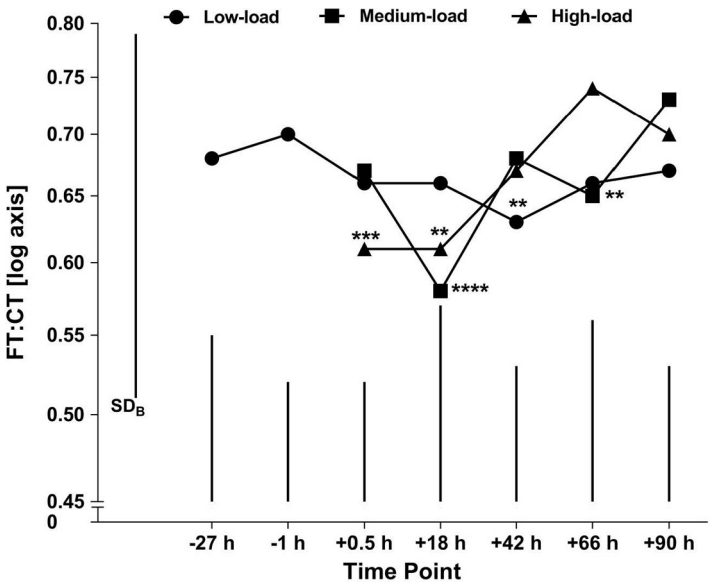

Figure 3 — Mean ratio of flight time to contraction time (FT:CT) from 27 hours pre to 90 hours post of low, medium, and high match loads. Mean values are displayed in log scale with the overall between-subjects SD shown by the bar labeled SD_B. The single bars underneath each time point represent the within-subject SD specific to that time point. Changes exceeding the effect size (0.2) with the qualitative descriptor of likely ** (75–<95%), very likely *** (95–99.5%), and most likely **** (>99.5%) are displayed.

Table 1 Difference in Magnitude of Change From Baseline of Jump Height, Ratio of Flight Time to Contraction Time (FT:CT), Cortisol, Testosterone, and Ratio of Testosterone to Cortisol (T:C) Between Levels of Match Load

	Match load	0.5 h post	18 h post	42 h post	66 h post
Jump height	low	2.4%, ±9.1%	0.6% ±5.2%	–2.6% ±5.3%	1.4%, ±5.8%
	medium	–9.9%, ±7.3%; ↓**	–6.6%, ±4.4%; ↓**	1.0%, ±4.8%	–4.5%, ±4.8%; ↓**
	high	–15.8 ±8.1%; ↓***	–8.7 ±5.3%; ↓***	3.0 ±6.9%	–1.4%, ±6.5%
FT:CT	low	–5.6%, ±6.6%; ↓*	–5.3%, ±9.9%	–9.6%, ±6.7%; ↓**	–7.8%, ±9.9%; ↓*
	medium	–5.0%, ±6.5%; ↓*	–17.0%, ±7.5%; ↓****	–3.6%, ±6.4%; ↓*	–9.2%, ±8.5%; ↓**
	high	–12.4%, ±6.6%; ↓***	–12.6%, ±10.3%; ↓**	–4.6%, ±8.6%	2.9%, ±13.0%
Cortisol (µg/dL)	low	54.6%, ±54.4%; ↑***	–0.0%, ±21.6%	–5.7%, ±19.9%	34.7%, ±27.0%; ↑***
	medium	165.4%, ±72.7%; ↑****	–14.2%, ±17.1%; ↓*	–30.7%, ±13.3%; ↓***	–1.0%, ±18.0%
	high	101.2%, ±62.6%; ↑****	2.9%, ±21.8%	–18.0%, ±16.9%; ↓**	–8.2%, ±17.1%
Testosterone (pg/mL)	low	17.0%, ±17.8%; ↑**	8.3%, ±15.2%	12.1%, ±14.9%; ↑**	21.6%, ±20.3%; ↑**
	medium	19.5%, ±15.0%; ↑***	–10.6%, ±11.4%; ↓**	–11.5%, ±10.7%; ↓**	–9.4%, ±13.3%; ↓*
	high	17.0%, ±16.1%; ↑**	5.8%, ±14.5%	0.8%, ±13.1%	–5.7%, ±14.3%
T:C	low	–27.2%, ±20.6%; ↓**	6.1%, ±18.0%	16.5%, ±18.0%; ↑**	–12.4%, ±13.4%; ↓**
	medium	–54.4%, ±10.0%; ↓****	5.1%, ±16.4%	29.0%, ±18.4%; ↑***	–7.0%, ±13.0%
	high	–42.5%, ±14.4%; ↓****	3.0%, ±17.1%	23.2%, ±18.6%; ↑***	4.1%, ±14.7%

Note: Values are presented as percentage change in mean, ± 90% CI; direction of response: positive ↑ and negative ↓.

*Possibly, **likely, ***very likely, and ****most likely chance of the true effect exceeding a small (0.2) effect size.

±7% reduction (moderate effect) in FT:CT at 0.5 hour post. At 18 hours, there was a most likely and likely reduction (moderate effects) in FT:CT in medium and high load, respectively.

Mean C, T, and T:C responses to the match-load tertiles are displayed in Figures 4, 5, and 6, respectively, with the change in mean from baseline in Table 1. Compared with 90 hours post, at 0.5 hour post there was a very likely (moderate effect) increase in C in low load, most likely increase in medium load (very large effect), and most likely increase in high load, with a large effect (range 55–165%). At 18 hours, the change in C was unclear. T displayed likely increases in low and high load and very likely increase with moderate effects in medium load (range 17–20%) at 0.5 hour post. There were varied responses thereafter. Low load resulted in a likely reduction (moderate effect) in T:C, with moderate and high load resulting in most likely reductions at 0.5 hour post, with very large and large effects, respectively. There were unclear changes in T:C in all load tertiles at 18 hours, with small (low load) and moderate (medium and high load) effects of an increased T:C at 42 hours.

Magnitude of Change From Baseline Between Levels of PlayerLoad

The magnitude of change from baseline between-matches load tertiles in all variables is presented in Table 2. At 0.5 hour, there was a moderate likely greater reduction (12%, ±11%) in jump height in medium load and large very likely greater reduction (18%, ±11%) in high load than in low load. There was a small likely greater reduction (8%, ±9%) in FT:CT in high load than in medium load at the same point. At 18 hours, there was a moderate likely greater reduction (12.4%, ±11.8%) in FT:CT in medium load than in low load. There was a moderate very likely increase (72%, ±77%) in C in medium load compared with low load at 0.5 hour post. Differences in the T response between load tertiles were variable, while the differences in T:C were predominantly unclear.

Discussion

The aim of this research was to determine the sensitivity of CMJ performance, T, C, and the T:C ratio to load accumulated during football match play. Elite-level football match load altered athletes' subsequent CMJ and hormonal response. Match load mediated the degree of change in jump height and FT:CT, with FT:CT more sensitive to match and subsequent training load. The results of this study highlight the need to consider multiple markers of recovery from match load.

The magnitude of reduction in jump height at 0.5 and 18 hours post appears to be modified by match load. For example, the change was unclear at these times in low load, yet very likely reduced with large effects in high load. Furthermore, at 0.5 hour, when compared with low load, there was a likely and very likely (moderate and large effects) lower jump height in medium and high load, respectively, but no clear difference between medium and high loads. Given this finding, it appears plausible that there is a minimum threshold match load that negatively affects CMJ height postmatch. Similarly a maximum threshold seems to exist beyond which there is limited additional clear impairment of jump performance. As the results indicate a dose-response relationship, this finding provides some support for the ecological validity of

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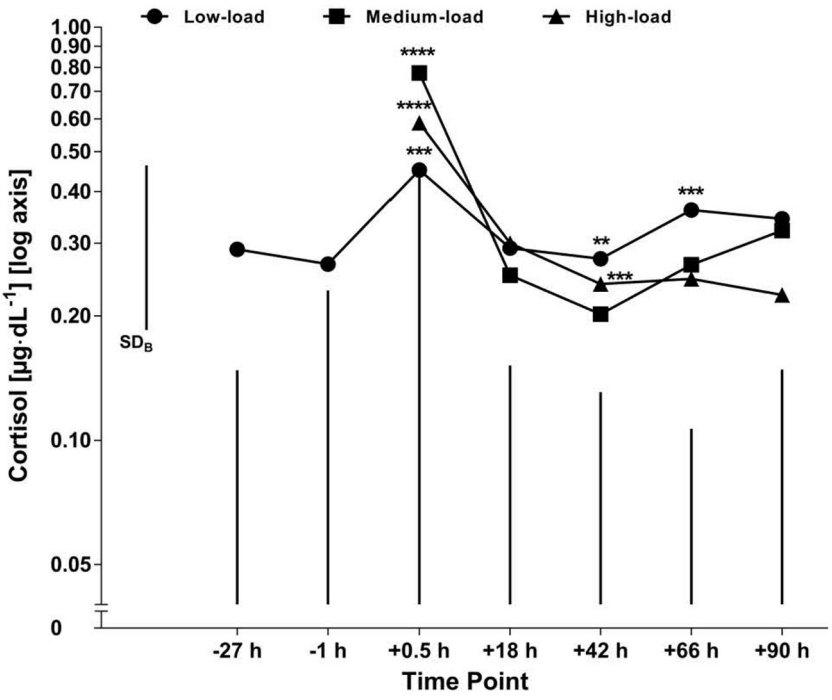

Figure 4 — Mean cortisol from 27 hours pre to 90 hours post of low, medium, and high match loads. Mean values are displayed in log scale with the overall between-subjects SD shown by the bar labeled SD_B. The single bars underneath each time point represent the within-subject SD specific to that time point. Changes exceeding the effect size (0.2) with the qualitative descriptor of likely ** (75–<95%), very likely *** (95–99.5%), and most likely **** (>99.5%) are displayed.

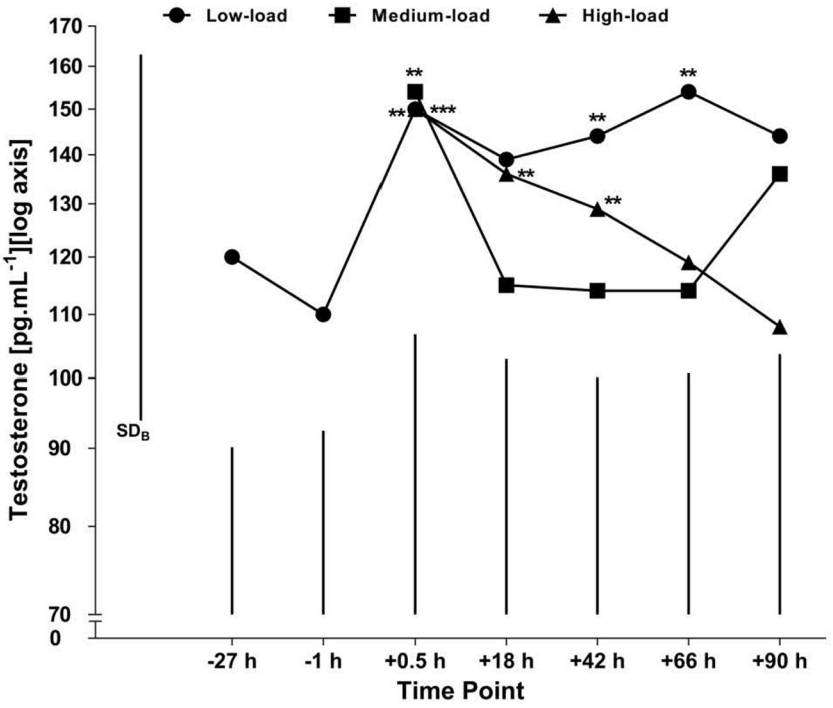

Figure 5 — Mean testosterone response from 27 hours pre to 90 hours post of low, medium, and high match loads. Mean values are displayed in log scale with the overall between-subjects SD shown by the bar labeled SD_B. The single bars underneath each time point represent the within-subject SD specific to that time point. Changes exceeding the effect size (0.2) with the qualitative descriptor of likely ** (75–<95%), very likely *** (95–99.5%), and most likely **** (>99.5%) are displayed.

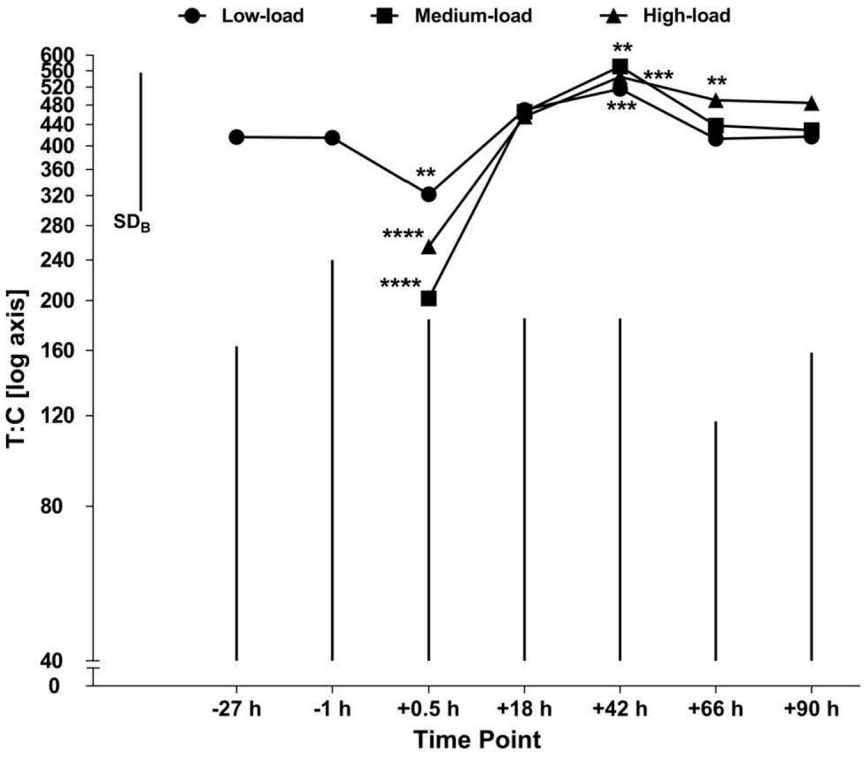

Figure 6 — Mean response of the ratio of testosterone to cortisol (T:C) from 27 hours pre to 90 hours post of low, medium, and high match loads. Mean values are displayed in log scale with the overall between-subjects SD shown by the bar labeled SD_B. The single bars underneath each time point represent the within-subject SD specific to that time point. Changes exceeding the effect size (0.2) with the qualitative descriptor of likely ** (75–<95%), very likely *** (95–99.5%), and most likely **** (>99.5%) are displayed.

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Table 2 Difference in Magnitude of Change Between Levels of Match Load of Jump Height, Ratio of Flight Time to Contraction Time (FT:CT), Cortisol, Testosterone, and Ratio of Testosterone to Cortisol (T:C)

	Match load	0.5 h post	18 h post	42 h post	66 h post
Jump height	M-L	–12.0%, ±10.7%; ↓**	–7.2%, ±6.7%; ↓**	3.6%, ±7.6%	–5.8%, ±7.1%; ↓**
FT:CT	M-L	0.6%, ±9.3%	–12.4%,±11.8%; ↓**	6.6%,±10.0%; ↑*	–1.6%, ±13.4%
Cortisol (µg/dL)	M-L	71.7%, ±76.6%; ↑***	–14.2%, ±24.6%	–26.4%, ±20.4%; ↓**	–26.5%, ±18.6%; ↓**
Testosterone (pg/mL)	M-L	2.2%, ±18.5%	–17.4%, ±14.2%; ↓**	–21.0%, ±12.7%; ↓***	–25.5%, ±15.0%; ↓***
T:C	M-L	–37.4%, ±22.4%; ↓***	–0.9%, ±22.2%	10.7%, ±22.5%	6.2%, ±20.5%
Jump height	H-M	–6.5%, ±11.8%	–2.2%, ±7.2%	2.0%, ±8.4%	3.2%, ±8.5%
FT:CT	H-M	–7.7%, ±8.7%; ↓**	5.3%, ±15.0%	–1.1%, ±10.4%	13.4%, ±17.2%; ↑**
Cortisol (µg/dL)	H-M	–24.2%, ±31.1%	19.9%, ±33.4%	18.3%, ±31.8%	–7.3%, ±21.8%
Testosterone (pg/mL)	H-M	–2.1%, ±16.1%	18.3%, ±19.8%; ↑**	13.9%, ±17.7%; ↑**	4.1%, ±19.5%
T:C	H-M	26.2%, ±41.3%	–2.0%, ±21.3%	–4.5%, ±18.7%	12.0%, ±20.1%
Jump height	H-L	–17.8%, ±10.8%; ↓***	–9.3%, ±7.0%; ↓**	5.7%, ±9.1%	–2.8%, ±8.5%
FT:CT	H-L	–7.2%, ±8.7%; ↓*	–7.7%, ±14.0%	5.5%, ±11.4%	11.6%, ±18.0%
Cortisol (µg/dL)	H-L	30.1, ±60.8%	2.9%, ±29.7	–13.0%, ±24.4%	–31.9%, ±17.35%; ↓***
Testosterone (pg/mL)	H-L	0.0%, ±18.6%	–2.3%, ±17.1%	–10.1, ±14.7%	–22.5%, ±15.9%; ↓***
T:C	H-L	–21.0%, ±29.5%	–2.9%, ±21.8%	5.7%, ±21.3%	18.9%, ±22.9%; ↑**

Note: Values are presented as percentage change in mean, ± 90% CI; direction of response: positive ↑ and negative ↓. Abbreviations: M-L: medium compared with low match load, H-M: high compared with medium match load, H-L: high compared with low match load.

*Possibly, **likely, ***very likely, and ****most likely chance of the true effect exceeding a small (0.2) effect size.

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jump height as a sensitive measure of an immediate neuromuscular response to football match play. While the mechanisms responsible for the reduction in postmatch jump height cannot be determined in this study, there are numerous possibilities.[12,17] These include central and peripheral factors including structural changes due to muscle microtrauma or reduced central drive.[11] At 42 hours there were unclear changes in jump height, while at 66 hours there was individual variability in the jump-height response to a given match load. This variability may be a function of the postmatch training load or related to individual athlete characteristics such as lower-body strength or fitness affecting athletes' tolerance to match loads.[14] Similar to the findings shown here, reduced jump height has been shown to last between 24 hours and 72 hours with variation according to level of football competition, timing of testing, and match versus simulated activity.[11,17,18]

Despite the support for jump height as a marker of recovery noted herein, the results are stronger in support of FT:CT. In contrast to the unclear change in jump height at 0.5 hour post in low load, FT:CT was possibly reduced in this group and medium load (although small effects were identified). High load resulted in a very likely (moderate effect) reduced FT:CT at 0.5 hour post. Although the impact of a possible reduction should not be overstated, it suggests a detection of a change in performance and provides different information about neuromuscular status than jump height. Compared with jump height at 18 hours post, there was a more likely reduction in FT:CT than jump height, supporting the contention that FT:CT is a more sensitive measure.[5,19] The likelihood of a reduction in FT:CT was also generally more certain than in jump height at 42 and 66 hours post, particularly in low load, where training occurred immediately after both 18- and 42-hour-postmatch jump testing. Thus, FT:CT may be more sensitive to acute load changes than jump height. The observed response may also be representative of the bimodal pattern of recovery from repetitive stretch-shortening-cycle exercise.[5,6] Regardless of mechanism, this finding adds weight to the concept that impairments to neuromuscular function result in a reorganization of jump strategy to maintain a similar output (eg, height).[6,18] In such a case, the outcome measure of flight time or jump height is relatively unaffected, but when considered relative to a measure of jump strategy (ie, contraction time), a decrement is revealed.[4,6,18]

In the comparison of the magnitude of change between load groups from 18 to 66 hours post, neither jump height nor FT:CT changed with more certainty. It is possible that the arbitrary division of external load used here was too narrow to detect clearly meaningful differences between the match-load tertiles. Support for this contention exists in the largely unclear differences in both jump height and FT:CT when comparing medium and high match loads. Beyond the 500-au threshold of low match load, it appears that individual variability (demonstrated by the wide confidence intervals) plays an important role. These individual differences may be related to position, tactics, and physical characteristics.[14] As PlayerLoad represents accelerations in 3 planes, it could be that some players have completed relatively more or less low-speed activity, high-speed running, sprinting, accelerating, decelerating, and changes of direction than others to produce the same absolute match load.[20] It could be that the distribution of activity profile has an impact on the FT:CT response, such that players involved in relatively more intermittent high-intensity activities may respond differently than those involved in a higher number of low-speed continuous actions. In addition, players in the same match-load group could be nearly 500 au apart, which may have affected the results.

The intensity of A League match play resulted in an immediate increase in salivary C across all load tertiles. As with CMJ variables, it is difficult to compare this result with other work, as the activity profile has not generally been accounted for. Intercollegiate male soccer players and rugby league players have had a more than 200% increase in C after match play,[15,21] while in Australian Rules football players, C increased immediately and 24 hours postmatch

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by 34% and 42%, respectively.[5] Although all load tertiles resulted in increased C postmatch, the increases were greater in medium and high load than in low. It appears that A League match play is of sufficient volume and intensity to cause an increased secretion of C.[15] Unlike the modified CMJ performance, the same dose-response relationship is not evident in C, which appears due to substantial individual variation. The reason for this variation is not obvious, but the biochemical response to team-sport exercise appears to be influenced by intermittent endurance fitness, and similar mechanisms may be at play up until 18 hours post.[14] In addition, C is influenced by psychological factors that may have played a role in the current results.[15,22] For example, higher postmatch values were observed in starters than in nonstarters in college football.[23] By 42 hours post, clear reductions in C are evident in the medium and high load (small and moderate effect, respectively), although the difference between these groups is unclear. Finally, the impact of circadian variation on the results due to some inconsistency in sample-collection time around matches cannot be ruled out.[21,22]

All load tertiles exhibited meaningful increases in T postmatch, with no clear difference in the magnitude of change between tertiles. Variable responses are common, with an average 15% increase postmatch in college football, an approximate 20% reduction after rugby match play, and no change after elite football match play.[15,21,24] At 18 hours post and beyond, the T response appears to be a function of substantial individual variability. The division of match load into the 3 broad groups and the potential impact of circadian variation as mentioned previously may have caused such variability.[21,22] Similar to the contention posed for FT:CT, it may be that despite a similar external load, players could have achieved this outcome with a markedly different activity profile, affecting individual T responses.

Low, medium, and high load resulted in moderate likely, very large, and large most likely reductions in T:C, respectively, immediately postmatch. The T:C ratio is said to be reflective of anabolic–catabolic balance, and the results of the current study suggest an immediate relatively catabolic response to A League match play.[5,15] Similar catabolic states existed after rugby union match play, where T:C was 2.5 times lower than on a rest day, and after first-division women's football match play, where there was a 32% reduction in T:C.[15,25] At 18 hours post, the change in T:C is uniformly trivial unclear. This result appears to contrast previous work in football where T:C was reduced for 48 hours compared with prematch.[12] However, the uncertainty described here may in fact be a function of the precise statistical analysis employed and more accurately reflect the true individual variability relative to match load.[26] Note that by 42 hours post there are small likely to moderate very likely increases in T:C, suggesting a return to an anabolic environment after the introduction of a training stimulus. It appears that there is a progressive change from a relatively catabolic to anabolic environment somewhere between immediately postmatch and 42 hours post. After rugby union match play, T:C changed from an immediate postmatch reduction to increased values in the following 4 days.[15] In a way similar to the reductions evident in FT:CT at 66 hours post, the clear reduction in T:C in low match load is potentially a function of training load between 42 and 66 hours post.[5]

Practical Applications

Football players accumulating a PlayerLoad of >500 au have reduced CMJ performance for at least 42 hours. The FT:CT is sensitive to both match load and the introduction of subsequent training load typical in a training week. Thus, FT:CT is the most useful CMJ variable for assessing recovery postmatch. Practitioners should consider assessing this variable from a CMJ at least 42 hours postmatch to help inform training prescription. The substantial individual variability and cost of analysis of hormonal response raise questions regarding the utility of regular monitoring of salivary hormones in A League football. Individual monitoring is recommended in a team environment.

Conclusion

External load accumulated during A League match play results in a reduction in CMJ performance and increased C and T concentrations. Reductions in jump height were greatest in medium and high loads at 0.5 and 18 hours postmatch, suggestive of a dose-response relationship. The FT:CT ratio detected an immediate alteration in jump performance associated with higher match loads and was reflective of altered movement strategy postmatch. This study identified wide individual variation, particularly with the hormonal response beyond immediately postmatch. Therefore, to fully understand the complexities of accounting for individual athlete recovery via neuromuscular and hormonal response to match play in the team environment, sophisticated modeling is needed.

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